Lakshdeep Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.65
Paradise Public School	Class XII (CBSE)	2023	97.40%
Sacred Heart School	Class X (ICSE)	2021	96.80%

SCHOLASTIC ACHIEVEMENTS

- **Ranked in the top 1% nationwide** among 36,000+ participants in the **National Standard Examination in Chemistry (NSEC)**, and secured **top 1% in the state** in the **National Standard Examination in Physics Olympiad (NSEP).**
- Peak Rating of 1346 on Codeforces.

EXPERIENCE

Microsoft Corp. - *SWE Internship*

Intelligent Incident Response Agent

- **Built and deployed an AI-powered incident-response agent** to identify communication gaps in operational workflows, automate escalation handling, and trigger targeted notifications to relevant owners.
- **Engineered a 1,400+ line deterministic decision engine** for conversational analysis, ownership resolution, and **multi-level escalation**, incorporating structured decision logic and **idempotent execution** for reliable automated actions.
- **Architected an end-to-end automation pipeline** using **Copilot Studio, Power Automate, and Azure Data Explorer**, integrating data processing, workflow orchestration, and **safety guardrails** for reliable production execution.

PROJECTS

Quant DSL [GitHub]

- Built **LOKUSQuant**, a domain-specific language for low-latency quantitative trading, implementing a compiler pipeline with **lexical analysis, parsing, semantic analysis, and code generation** using C++/LLVM/MLIR.
- Designed a **domain-aware type system and semantic analyser** for financial primitives, enforcing type safety and validating strategy declarations, expressions, immutability, and scope-related constraints at compile time.
- Implemented compiler-driven **Signal Dependency Graph (SDG) optimisation** to identify shared computations across trading signals and strategies, enabling incremental evaluation and eliminating redundant per-tick computation.

RISC-V ISA Cache Simulator [Hosted Site] [GitHub]

- Developed a full-stack web application with **React, Express.js** and **Socket.IO** integrating **C++** with **OOP** principles to **simulate RISC-V assembly** code execution and cache behavior.
- Implemented core **RISC-V** instructions, enabling analysis of **memory access** patterns including hits, misses, and access times.
- Designed an interactive UI platform with an editor to write and edit code and multiuser support using Socket.IO.

RBAC Microfrontend Platform [GitHub]

- Built a **microservices-based RBAC platform** with **Node.js, Express.js** and **MongoDB** for centralized authentication, role and permission management.
- Developed a centralized **JWT-based authorization engine** supporting fine-grained access control across **microservices and microfrontends**.
- Designed independently deployable **React/Vite microfrontends** using **Module Federation** with registry-driven service integration and role-based access control.

TLS Security Scanner [GitHub]

- Developed a modular **TLS security scanner** in **Python** to analyze server configurations for **certificate, protocol, cipher, feature, and vulnerability** issues.
- Implemented extensible **check registry and scanning pipeline** with configurable checks, structured findings, severity classification, error handling, and connection context management.
- Built a **multiprocessing-based parallel scanner** with worker queues and configurable timeouts to scan multiple TLS endpoints concurrently, alongside benchmark tooling for evaluating scalability.

Vehicle Rental System (REST API + DBMS) [GitHub]

- Built a full-stack Vehicle Rental System using **React, Express.js,** and **MySQL**, featuring a RESTful API and structured data management.
- Implemented secure authentication and **role-based authorization**, ensuring data consistency and multi-user concurrency using **transactions, triggers**, and **stored procedures** in SQL.

POSITIONS OF RESPONSIBILITY

Core Member at Lambda (Software Development Club, IIT Hyderabad) **:**

- Contributed to the development of the **IITH Dashboard**, implementing backend **Gin routes** for modules like Lost & Found, mess menu, and bus timings to enhance campus-wide accessibility.

Teaching Assistant for **Introduction to Programming** Course (ID1063) **:**

- **Mentored** first-year students in **C programming** during labs and exams and assisted in paper evaluation.

Teaching Assistant for **Computer Architecture** (CS2323) **:**

- Mentored sophomores on **RISC-V ISA, pipelining, and core computer architecture concepts**, and assisted with the evaluation of assignments and examinations.

SKILLS

- **Programming Languages :** C, C++, Python, Javascript, Java, SQL, Go, Bash, RISC-V ASM.
- **Frameworks/Libraries:** React, Next.js, Gin, Django, Bootstrap, Tailwind CSS, Cytoscape.js, NumPy, Pandas, Matplotlib, Socket.IO, Cytoscape.js, pygame.
- **Databases:** MySQL, PostgreSQL, MongoDB.
- **Tools:** Git, Docker, VSCode, Vim, Postman, LaTeX, Makefile.

RELEVANT COURSES

Computer Science: Foundations of Machine Learning, Deep Learning, Fraud Detection and Analytics, Software Engineering, Compilers, Computer Networks, DBMS, Parallel and Concurrent Programming, Operating Systems, Computer Architecture, Algorithms, Data Structures, Approximation Algorithms, Theory of Computation.